

STAT 654 - More General Regression

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More General Regression Analyses

Eventually we will study the case where we have more than one predictor/co-variate/independent variable.

But initially we discuss a way of handling the one-variable case when a straight line fit (i.e. a linear regression model) isn't adequate.

Simulated Data

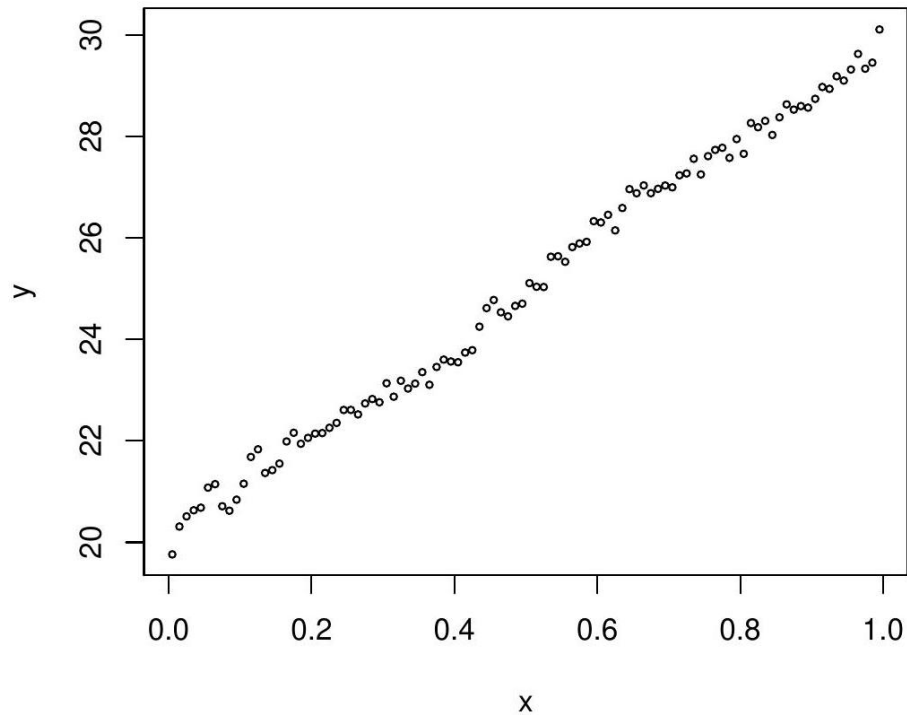
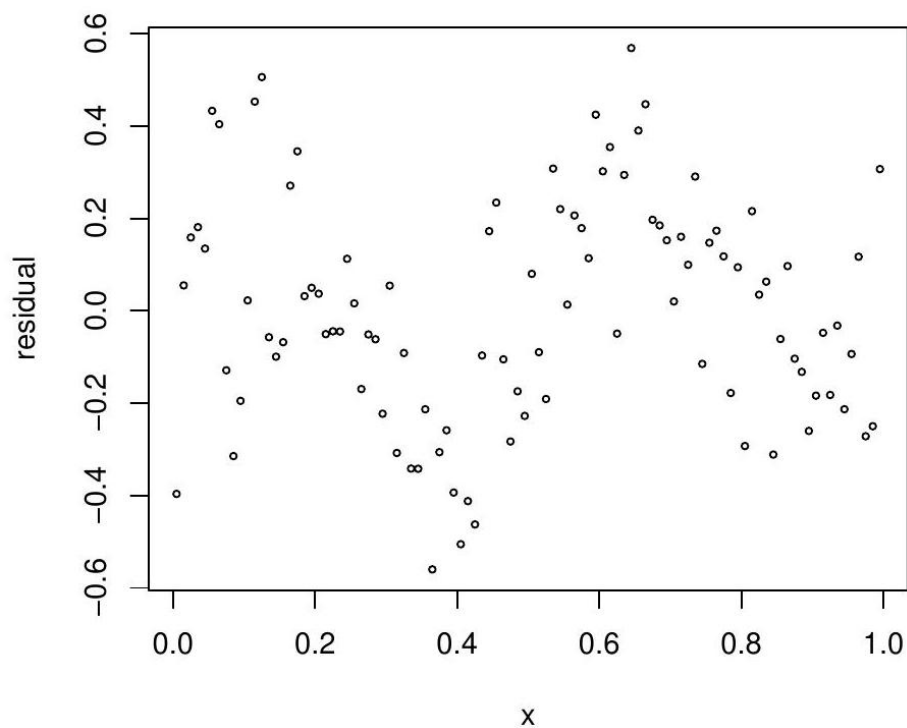


Figure 1: Spotting nonlinearity via residuals

Residuals from a straight line fitted to data on previous page



The data were generated from the following model:

$$Y_j = 20 + 10x_j + 0.25 \sin(4\pi x_j) + \epsilon_j,$$

$j = 1, \dots, 100$, where each ϵ_j is normally distributed with mean 0 and std. deviation 0.2.

Previously our model was:

$$Y = \beta_0 + \beta_1 x + \epsilon$$

More generally, we could assume the following:

$$Y = f(x) + \epsilon$$

where f is an arbitrary function.

However, when doing a statistical analysis we must limit the set of candidate models. Otherwise there would be too many parameters to estimate.

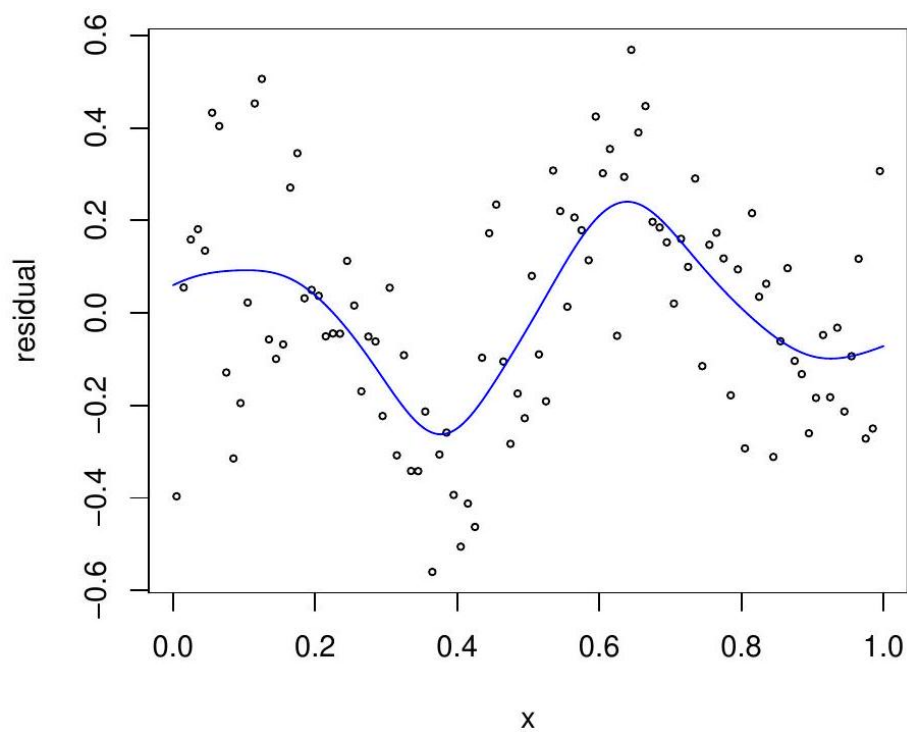


Figure 2: Residuals and smoothed residuals

Principle of parsimony or **Occam's razor**:

The best explanation of a phenomenon is the simplest one that fits the facts.

Statistical analog of Occam's razor:

Use the simplest model, i.e., the one with fewest parameters, that still fits the data adequately.

A good way to obtain a “flexible” class of models while limiting the number of parameters is to use polynomials (i.e. $f(x)$ is a polynomial).

Taylor's theorem tells us that a “smooth” function can be approximated arbitrarily well by a polynomial of sufficiently high degree.

We will assume that:

$$Y = \beta_0 + \beta_1 x + \beta_2 x^2 + \cdots + \beta_k x^k + \epsilon.$$

As before $\epsilon \sim N(0, \sigma^2)$.

Given the data $(x_1, y_1), \dots, (x_n, y_n)$, we can estimate the polynomial coefficients using least squares (the principle stays same as before).

$$f(b_0, b_1, \dots, b_k) = \sum_{i=1}^n \left(y_i - \sum_{j=0}^k b_j x_i^j \right)^2$$

Choose $b_0, \dots, b_k$ to minimize $f(b_0, b_1, \dots, b_k)$.

$$\frac{\partial f(b_0, \dots, b_k)}{\partial b_j} = 0, \quad j = 0, 1, \dots, k$$

This leads to the set of equations on the next page which are the so-called **normal equations**.

$$\begin{aligned} \sum_{j=0}^k b_j \sum_{i=1}^n x_i^j &= \sum_{i=1}^n y_i \\ \sum_{j=0}^k b_j \sum_{i=1}^n x_i^{j+1} &= \sum_{i=1}^n x_i y_i \\ &\vdots \\ \sum_{j=0}^k b_j \sum_{i=1}^n x_i^{j+k} &= \sum_{i=1}^n x_i^k y_i \end{aligned}$$

These have the form

$$a_{\ell 0} b_0 + a_{\ell 1} b_1 + \dots + a_{\ell k} b_k = c_\ell, \quad \ell = 0, \dots, k$$

and are thus linear equations.

Linear equations are easy to solve, which is a big plus for polynomial regression.

We may derive the solution by writing the equations in matrix form.

Notation: If $\mathbf{A}$ is a $p \times q$ matrix, then $\mathbf{A}^T$ denotes the transpose of $\mathbf{A}$. $\mathbf{A}^T$ is a $q \times p$ matrix whose columns are the same as the rows of $\mathbf{A}$.

Define the following matrices:

$$\mathbf{b} = \begin{bmatrix} b_0 & b_1 & \cdots & b_k \end{bmatrix}^T$$

$$\mathbf{X} = \begin{bmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^k \\ 1 & x_2 & x_2^2 & \cdots & x_2^k \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^k \end{bmatrix}$$

$$\mathbf{y} = \begin{bmatrix} y_1 & y_2 & \cdots & y_n \end{bmatrix}^T$$

The normal equations in matrix form are:

$$\mathbf{X}^T \mathbf{X} \mathbf{b} = \mathbf{X}^T \mathbf{y}$$

Assuming that $\mathbf{X}^T \mathbf{X}$ is invertible, the soln. is:

$$\mathbf{b} = \left(\mathbf{X}^T \mathbf{X} \right)^{-1} \mathbf{X}^T \mathbf{y} = \hat{\boldsymbol{\beta}}$$

The components of $\hat{\boldsymbol{\beta}}$ are the least squares estimates of $\beta_0, \beta_1, \dots, \beta_k$.

Estimating σ^2

Predicted values:

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i + \cdots + \hat{\beta}_k x_i^k, \quad i = 1, \dots, n$$

Residuals:

$$e_i = y_i - \hat{y}_i, \quad i = 1, \dots, n$$

$$SSE = \sum_{i=1}^n e_i^2$$

We estimate σ^2 by:

$$\hat{\sigma}^2 = MSE = \frac{SSE}{n - k - 1}$$

As in the straight line case: $SST = SSR + SSE$, with $SST = \sum_{i=1}^n (y_i - \bar{y})^2$, $SSR = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$

$$\text{and } R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

R^2 is the fraction of the total variation in the y_i 's explained by the polynomial model.

Using *R* to fit polynomials

There are two ways to fit, say, a third degree polynomial:

- `lm(y ~ poly(x, 3, raw = T))`, or equivalently

$$\text{lm}(y \sim x + I(x^2) + I(x^3))$$

Both methods produce exactly the same predicted values and residuals.

The commands:

```
fit=lm(y~poly(x,3,raw=T))
summary(lm(y~poly(x,3,raw=T)))
```

provide us with similar looking output as in the straight line case. Actually all prior *R* commands continue to apply.

Example: Fitting different degree polynomials to a data set

Data were simulated from the model

$$Y_i = 11.8 + 28x_i - 70x_i^2 + 50x_i^3 + \epsilon_i,$$

where $x_i = (i - 1/2)/30, i = 1, \dots, 30$, and $\epsilon_i \sim N(0, 1)$.

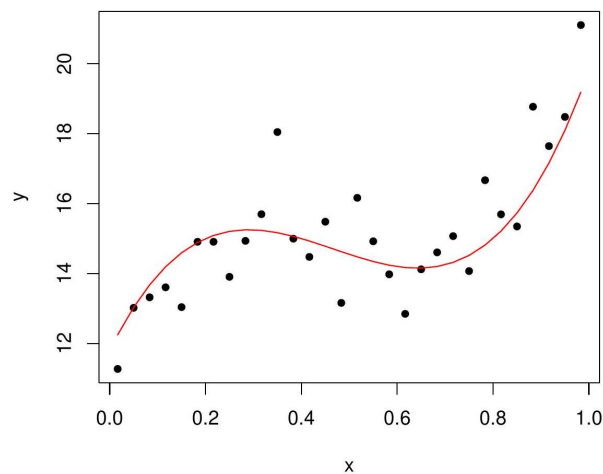


Figure 3: Simulated data and **true** cubic polynomial

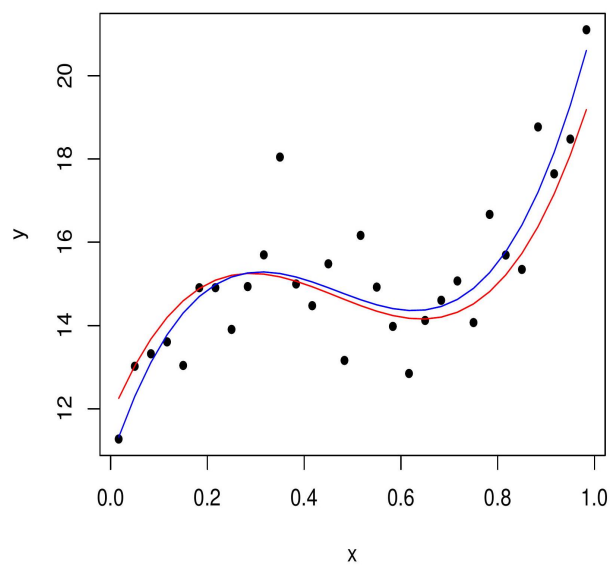


Figure 4: Red: true curve. Blue: cubic fit

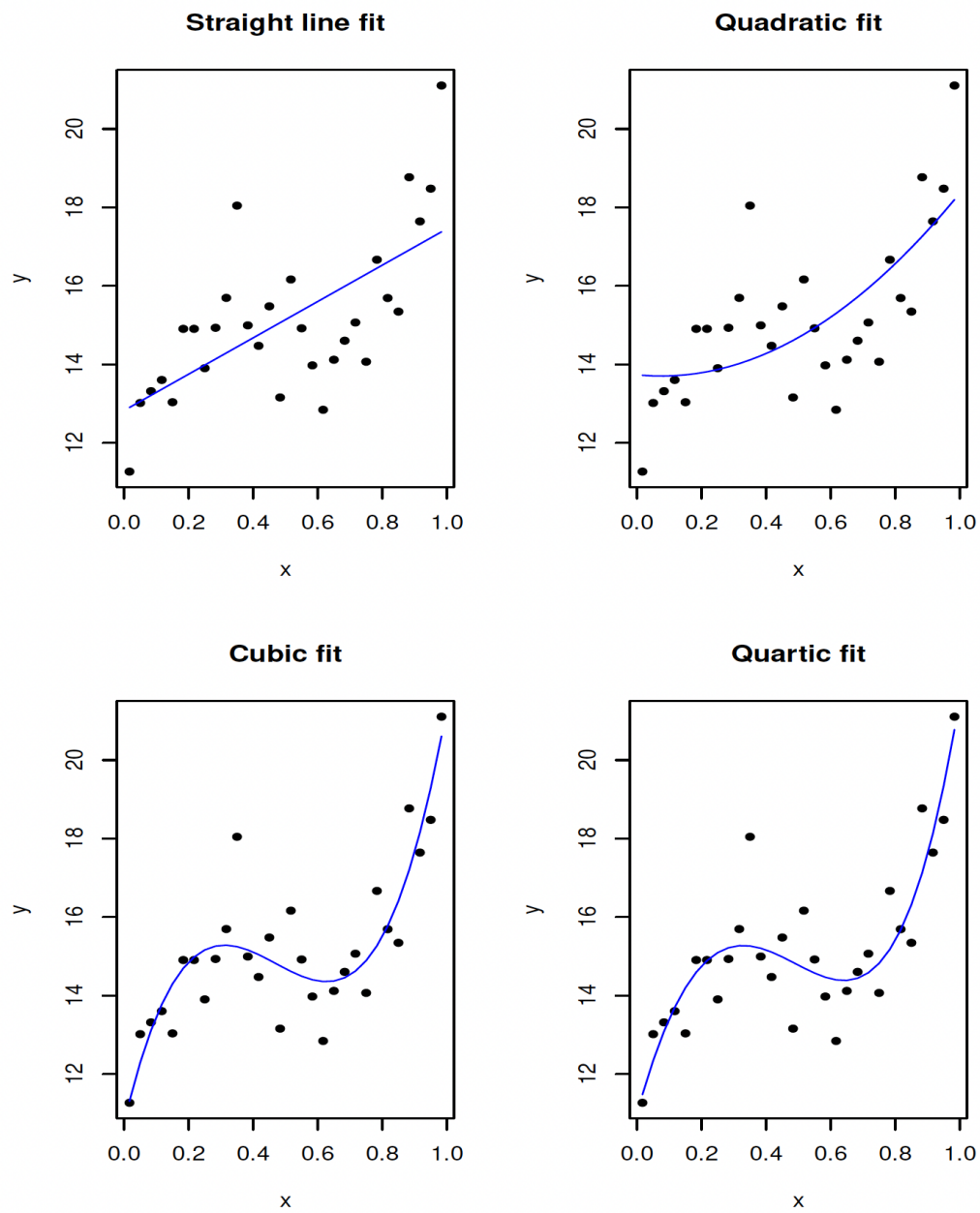


Figure 5: Fitting with degrees 1,2,3 and 4.

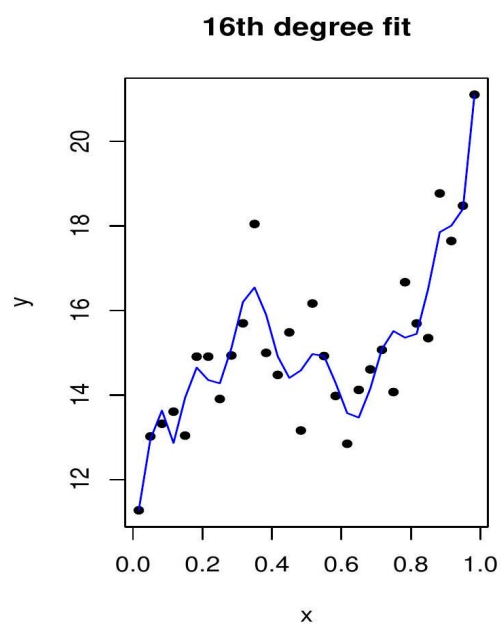
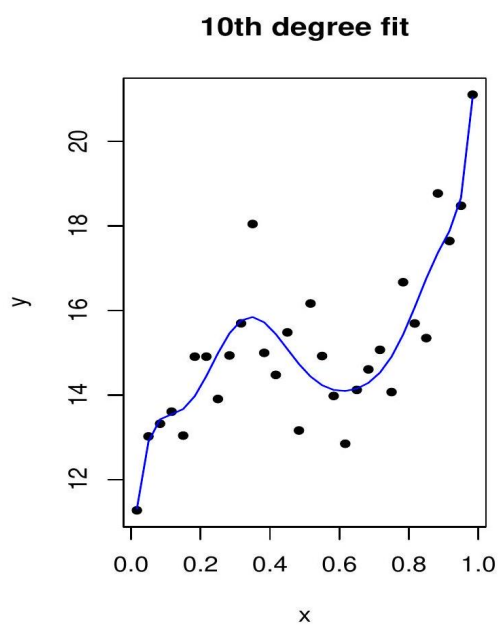
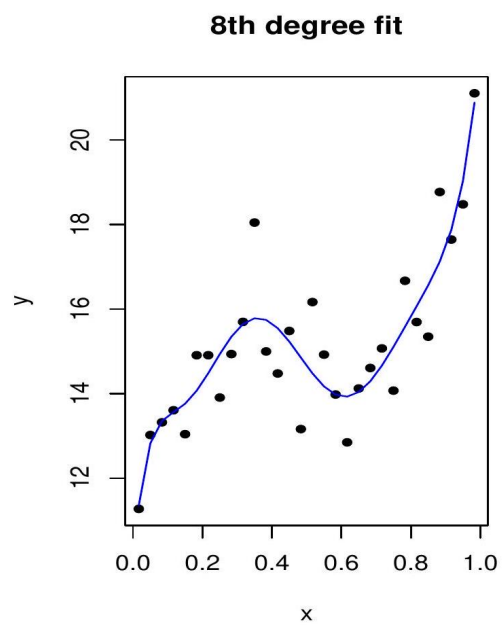
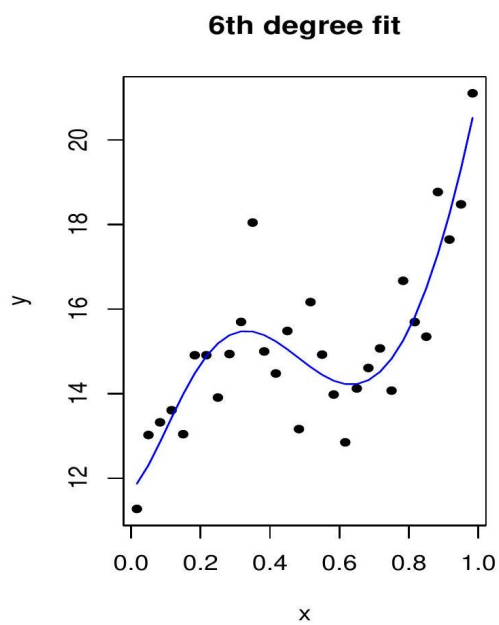
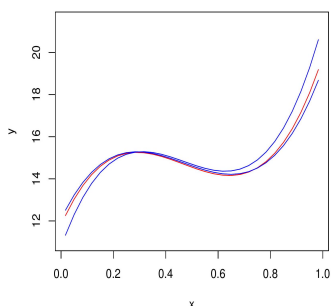
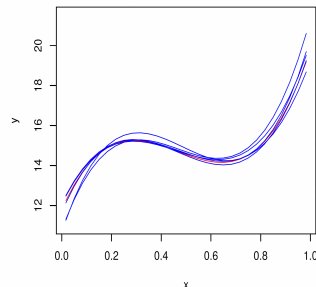


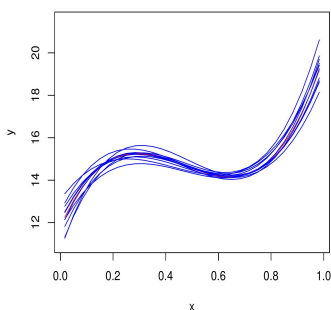
Figure 6: Fitting with degrees 6,8,10 and 16.



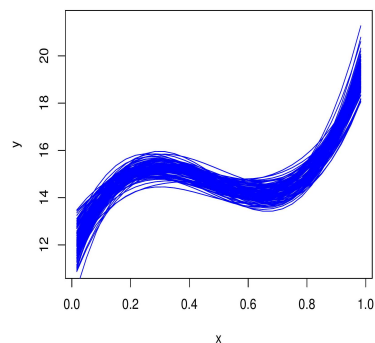
(a) Fitted curves from two different data sets.



(b) Fitted curves from five different data sets.



(c) Fitted curves from two different data sets.



(d) Fitted curves from five different data sets.

Figure 7: Fitted curves from different data sets

Question: How do you do these with real data?
 Answer: Bootstrap.

Adjusted R^2

When using polynomials to estimate a regression function, we won't always know beforehand (i.e., before data collection) what a good polynomial degree is.

One might be tempted to say "Use the degree that gives the highest R^2 value."

This turns out to be unreliable since R^2 is guaranteed to be an increasing function of polynomial degree.

Let R_k^2 be the R^2 value when a k th degree polynomial is fitted to our data. Then

$$R_1^2 < R_2^2 < \dots < R_{n-2}^2 < R_{n-1}^2 = 1.$$

So, maximizing R^2 will not help us to act in accord with Occam's razor.
 One solution: Use adjusted R^2 , denoted $R^2_{\text{adj},k}$.

$$\begin{aligned} R^2_{\text{adj},k} &= \frac{(n-1)R_k^2 - k}{n-1-k} \\ &= R_k^2 - \frac{k}{(n-1-k)} (1 - R_k^2). \end{aligned}$$

So $R^2_{\text{adj},k}$ penalizes R_k^2 and the penalty increases with k . Choose k to maximize $R^2_{\text{adj},k}$. The higher degree polynomials are penalized more.

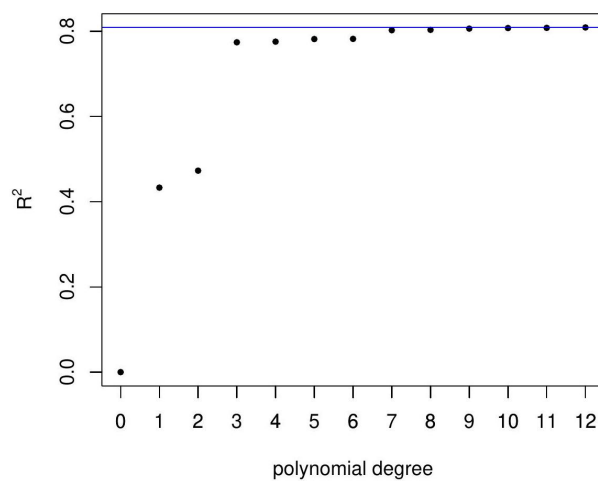


Figure 8: Unadjusted R^2 for simulated data.

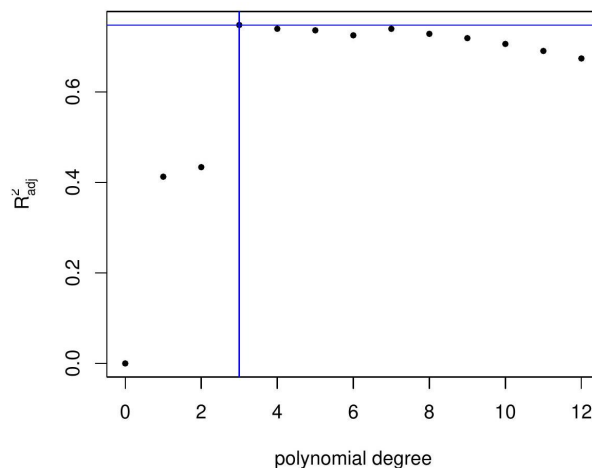


Figure 9: Adjusted R^2 for simulated data.

So, R_{adj}^2 worked very well in this example since it is maximized at 3, which matches the true polynomial degree.

Two other methods of selecting polynomial degree are called **AIC** (Akaike Information Criteria) and **BIC** (Bayes Information Criteria).

Using R to get AIC, BIC and Adjusted R^2

Suppose you use the command:

```
fit = lm(y ~poly(x, 3, raw=T))
```

to fit a third degree polynomial model. You may obtain the values of AIC, BIC and R_{adj}^2 as follows:

- AIC: `AIC(fit)`
- BIC: `BIC(fit)`
- Adjusted R^2 : `summary(fit) $ adj`

Choose polynomial degree to minimize AIC (and likewise for BIC).

How well do the three methods work?

I generated 1000 data sets from the third degree polynomial model.

For each of the 1000 data sets, I determined the degree that was selected by AIC, BIC and adjusted R^2 , where degrees from 1 through 12 were considered.

Percentage of 1000 cases where selectors chose degree k

k	AIC	BIC	R^2_{adj}
1	0.1	0.2	0
2	0	0	0
3	52.5	84.0	23.1
4	9.1	6.6	8.1
5	6.7	4.2	7.1
6	4.5	1.9	6.6
7	3.1	0.5	5.5
8	3.9	0.5	7.4
9	3.9	0.7	8.6
10	3.7	0.4	7.2
11	4.8	0.5	9.2
12	7.7	0.5	17.2

It's clear that BIC did the best job of selecting the correct degree, while AIC was second best.

Adjusted R^2 was a poor third. It has a stronger tendency to greatly overestimate the right degree than do AIC and BIC.

What happens if n is increased here? All results get more precise and with higher chances of choosing the correct degree.

FYI, one accepted definition for AIC and BIC:

Let SSE_k denote the SSE for the model fitted with the k^{th} degree polynomial. Then, AIC and BIC for this k^{th} degree model are given by:

$$\begin{aligned} \text{AIC}_k &= n \log(SSE_k) + 2k \quad \text{and} \\ \text{BIC}_k &= n \log(SSE_k) + k \log(n). \end{aligned}$$

Note that BIC penalizes more than AIC, and hence, tends to select smaller/simpler models.

Inference problems

Suppose we want to test a hypothesis about or construct a confidence interval for one of the β_i s in our polynomial regression model.

For example, consider the model

$$Y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \beta_3 x_i^3 + \epsilon_i.$$

This says that the polynomial degree is no more than 3 , but possibly less than 3.

Interesting hypotheses for the cubic model are:

$$H_0 : \beta_3 = 0 \quad \text{vs.} \quad H_a : \beta_3 \neq 0$$

H_0 says we have a quadratic model.

H_a says we have a cubic model.

The standard error of $\hat{\beta}_i$ is $\sigma\sqrt{c_i + 1}$, where for any $j = 1, \dots, k + 1$,

$$c_j = j \text{ th element on the} \\ \text{diagonal of } (\mathbf{X}^T \mathbf{X})^{-1}$$

When $\epsilon_1, \dots, \epsilon_n$ are a random sample from the normal distribution with mean 0 and var. σ^2 ,

$$\frac{\hat{\beta}_i - \beta_i}{\hat{\sigma}\sqrt{c_{i+1}}} \sim t_{n-k-1}$$

So, a $(1 - \alpha)100\%$ confidence interval for β_i is:

$$\hat{\beta}_i \pm t_{n-k-1; \alpha/2} \hat{\sigma} \sqrt{c_{i+1}}$$

Suppose we want to estimate the mean of Y at x -value x_0 , or predict Y corresponding to an x -value of x_0 .

The point estimate of the mean of Y at x_0 is:

$$\hat{\beta}_0 + \hat{\beta}_1 x_0 + \dots + \hat{\beta}_k x_0^k = \hat{\mu}(x_0).$$

$(1 - \alpha)100\%$ confidence interval for the mean response, i.e. mean of Y , at x_0 :

$$\hat{\mu}(x_0) \pm t_{n-k-1; \alpha/2} \hat{\sigma} \sqrt{x^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}} \\ \text{where } \mathbf{x}^T = (1 \quad x_0 \quad x_0^2 \quad \dots \quad x_0^k)$$

A $(1 - \alpha)100\%$ prediction interval for Y at x_0 :

$$\hat{\mu}(x_0) \pm t_{n-k-1; \alpha/2} \hat{\sigma} \sqrt{1 + \mathbf{x}^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}}$$

Inference for polynomial regression via R

All the commands we talked about in the straight line case (i.e. for simple linear regression) work exactly the same in polynomial regression.

Suppose we have fitted a third degree polynomial using the following command:

```
fit=lm(y~poly(x,3,raw=T))
```

Then the commands `summary`, `confint` and `predict` apply just as before.

Non-polynomial models

Suppose

$$Y_i = f(x_i; \theta_1, \dots, \theta_k) + \epsilon_i, \quad i = 1, \dots, n$$

where the ϵ_i 's satisfy our usual assumptions and the regression function $f(x; \theta_1, \dots, \theta_k)$ is known except for the parameters $\theta_1, \dots, \theta_k$.

For example,

$$f(x; \theta_1, \theta_2, \theta_3) = \exp(\theta_1 + \theta_2 x + \theta_3 x^2)$$

or

$$f(x; \theta_1, \theta_2) = \theta_1 \cos(2\pi\theta_2 x)$$

The choice of functional form would be based on knowledge of the particular problem.

Least squares can still be used to estimate the model parameters $\theta_1, \dots, \theta_k$.

Least squares estimates are those values $\hat{\theta}_1, \dots, \hat{\theta}_k$ that minimize

$$\sum_{i=1}^n [y_i - f(x_i; a_1, \dots, a_k)]^2$$

with respect to $a_1, \dots, a_k$.

Finding least squares estimates is often more difficult for non-polynomial models. Usually the equations to be solved are non-linear.

Transforming variables

Sometimes a model can be turned into a simple linear model by transforming x and/or y .

For example, suppose

$$Y = \alpha e^{\beta x} \epsilon,$$

where ϵ is a positive-valued random variable. Here the error is multiplicative rather than additive like it was before.

Taking the log of Y gives

$$\log Y = \log \alpha + \beta x + \log \epsilon$$

Now, defining

$$Y' = \log Y, \quad \beta_0 = \log \alpha, \quad \beta_1 = \beta$$

and $\epsilon' = \log \epsilon$, we get

$$Y' = \beta_0 + \beta_1 x + \epsilon'$$

which is just our old straight line model.

Similarly, if $Y = \alpha x^\beta \epsilon (x > 0)$,

$$\log Y = \log \alpha + \beta \log x + \log \epsilon$$

Taking $x' = \log x$, again we have our familiar straight line model.

For $Y = \alpha x^\beta \epsilon$ we have

$$\text{Var}(Y) = \alpha^2 x^{2\beta} \text{Var}(\epsilon)$$

and so even if ϵ has the same variance for all x , $\text{Var}(Y)$ will increase as x increases. On the other hand,

$$\text{Var}(\log Y) = \text{Var}(\log \epsilon)$$

which is the same for all x if the probability distribution of ϵ doesn't depend on x .

When the variance of the response appears to be increasing or decreasing with x , this is a tipoff that taking logs may be advisable.